The Great Outdoors, Indoors

An Evaluation of Green Spaces and Housing Prices in Pittsburgh, PA

By Will McCullough

Presented in Partial Fulfillment of the
Requirements of Senior Independent Study for the Department of
Economics at the College of Wooster

Supervised by
Dr. Moses Luri
Department of Economics

Spring 2021

Abstract

Green spaces, or public parks, playgrounds, and athletic fields, are a community hub. Many studies have highlighted the benefits of green space on surrounding neighborhoods. However, comparatively little research has assessed the relationship between park proximity and housing sales price. In Pittsburgh, Pennsylvania, city officials have shown a renewed interest in increasing park access and quality, although no economic literature has examined role of green space in the housing consumption decision in the city. I develop a theory of green space preference in a consumer maximization framework to hypothesize that home prices will increase with proximity to green space. I then use a Spatial Auto-Regressive Moving Average (SARMA) model to estimate the impact of green space proximity on single family home sales prices in Allegheny County, Pennsylvania. My results show that house sales prices will decrease by 2.34% with each additional mile from green space. This indicates a price premium on access to green space, further exacerbating inequalities between high and low socioeconomic status neighborhoods. This research highlights the need for additional research into environmental justice topics in Pittsburgh.